1 Related designations

The following tables of related designations are intended as a guide to alloy correspondence on the basis of chemical composition. The equivalents should not be taken as exact, and in all cases of doubt the relevant national specification or standard should be consulted. The tables do not represent an exhaustive list of all alloys available; the references listed at the end of the chapter are more complete sources.

In the case of the United Kingdom, France, and Germany, the tables refer to designations and standards that have recently been superseded by European (EN) and international (ISO) standards. The older standards have been referenced because the alloy designations are still in common use, and because the new standards use in some cases several different designations for chemically identical alloys, depending on the product form. A list of all designations is beyond the scope of this book; the references should be consulted for further details.

Table 1.1 lists designations for steels, with subsections for steels of different types. Designations for wrought aluminium alloys are listed in Table 1.2; cast aluminium alloys are shown in Table 22.2. Table 1.3 gives related copper alloy designations, subdivided into coppers, brasses, bronzes, and nickel silvers. Magnesium alloys cast and wrought are in Table 1.4. Related designations for nickel and titanium alloys are listed in Sections 22.5 and 22.6, respectively.

Unified number designations—UNS designations are five-digit numbers prefixed by a letter that characterises the alloy system as shown below.

UNS Letter	Designation[1]
A	Aluminium and aluminium alloys
B	Copper and copper alloys
D	Specified mechanical properties steels
E	Rare earth and rare earth like metals and alloys
F	Cast irons and cast steels
G	AISI and SAE carbon and alloy steels
H	AISI H-steels
J	Cast steels (except tool steels)
K	Miscellaneous steels and ferrous alloys
L	Low melting metals and alloys
M	Miscellaneous nonferrous metals and alloys
N	Nickel and nickel alloys
P	Precious metals and alloys
R	Reactive and refractory metals and alloys
S	Heat and corrosion resistant (stainless) steels
T	Tool steels
W	Welding filler metals

Table 1.1 RELATED DESIGNATIONS FOR STEELS

Nominal composition	*USA AISI/SAE (UNS)*	*UK BS 970 (En)*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Sweden SIS*	*France AFNOR*	*China GB*
1.1.1 Carbon steels								
C 0.06 Mn 0.35	1006 (G10060)	030A04	Ck7 (1.1009)	—	08Fkp	14 1147	XC6FF	05F
C 0.08/0.13 Mn 0.3/0.6	1010 (G10100)	045M10 (32A)	Ck10 (1.1121)	S9Ck	10kp	14 1311	CC10, AF34	10F
C 0.13/0.18 Mn 0.3/0.6	1015 (G10150)	050A15	Ck15 (1.1141)	S15Ck	15kp	14 1370	XC15	15F
C 0.17/0.23 Mn 0.3/0.6	1020 (G10200)	050A20 (2C, 2D)	Ck22 (1.1151)	S20C	20kp	14 1450	XC18, C20	20F
C 0.22/0.28 Mn 0.3/0.6	1025 (G10250)	080A25	Ck25 (1.1158)	S25C	M26	—	XC25	1025
C 0.27/0.34 Mn 0.6/0.9	1030 (G10300)	080A30 (5B)	Ck30 (1.1178)	S30C	30G	—	XC32	30
C 0.31/0.38 Mn 0.6/0.9	1035 (G10350)	080A35 (8A)	Ck34 (1.1173)	S35C	35	14 1572	XC35, C35	35
C 0.36/0.44 Mn 0.6/0.9	1040 (G10400)	080A40 (8C)	Ck40 (1.1186)	S40C	40	—	AF60, C40	40
C 0.42/0.50 Mn 0.6/0.9	1045 (G10450)	080M46	Ck45 (1.1191)	S45C	45	14 1672	XC45	45
C 0.47/0.55 Mn 0.6/0.9	1050 (G10500)	080M50	Ck50 (1.1206)	S50C	50	14 1674	XC50	50
C 0.55/0.65 Mn 0.6/0.9	1060 (G10600)	080A62 (43D)	Ck60 (1.1221)	S58C	60, 60G	14 1678	XC60	60
C 0.65/0.76 Mn 0.6/0.9	1070 (G10700)	080A72	Ck67 (1.1231)	—	70, 70G	14 1770	XC70	70
C 0.74/0.88 Mn 0.6/0.9	1080 (G10800)	080A83	80Mn4 (1.1259)	—	80	14 1774	XC80	80
C 0.90/1.04 Mn 0.3/0.5	1095 (G10950)	060A99	Ck101 (1.1274)	SUP 4	—	14 1870	XC100	—

1.1.2 Carbon-manganese and free-cutting steels

C 0.10/0.16 Mn 1.1/1.4	1513 (G15130)	130M15 (201)	12Mn6 (1.0496)	—	—	—	12M5	—
C 0.15/0.21 Mn 1.1/1.4	1518 (G15180)	120M19	21Mn4 (1.0469)	SMnC420	20GLS	14 2135	20M5	18MnSi
C 0.18/0.24 Mn 1.1/1.4	1522 (G15220)	150M19 (14A, 14B)	20Mn6 (1.1169)	SMn21	20G2	14 2165	20M5	18MnVB
C 0.30/0.38 Mn 1.2/1.55	1536 (G15360)	120M36 (15B)	36Mn5 (1.1167)	SMn1	35GL	14 3562	32M5	35Mn2
C 0.36/0.45 Mn 1.35/1.65	1541 (G15410)	150M40	36Mn6 (1.1127)	SMn21	40G2	14 2120	40M5	40Mn2
C 0.35/0.43 Mn 1.35/1.65 S 0.13/0.20	1139 (G11390)	216M36 (15AM)	35S20 (1.0726)	SUM41	A40G	14 1957	35MF4	Y40Mn
C 0.40/0.48 Mn 1.35/1.65 S 0.24/0.33	1144 (G11440)	226M44	45S20 (1.0727)	SUM43	A40	14 1973	45MF6	—
C 0.12 Mn 0.7/1.0 S 0.16/0.23	1212 (G12120)	—	9S20 (1.0711)	SUM21	A11	—	12MF4	—
C 0.13 Mn 0.7/1.0 S 0.24/0.33	1213 (G12130)	230M07	9SMn28 (1.0715)	SUM22	—	14 1912	S250	—

1.1.3 Alloy steels

C 0.1/0.2 Mn 0.3/0.6 Cr 0.6/0.95 Ni 2.75/3.25	3415	655M13 (36A)	14NiCr14 (1.5752)	SNC22H	12ChHN3A	—	12NC15	12CrNi3
C 0.23/0.28 Mn 0.7/0.9 Cr 0.4/0.6 Mo 0.2/0.3	4125	708A25	25CrMo4 (1.7218)	SCCrM1	25ChGM	14 2225	25CD45	—

(*continued*)

Table 1.1 RELATED DESIGNATIONS FOR STEELS—*continued*

Nominal composition	*USA AISI/SAE (UNS)*	*UK BS 970 (En)*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Sweden SIS*	*France AFNOR*	*China GB*
C 0.35/0.40 Mn 0.7/0.9 Mo 0.2/0.3	4037	605H37	GS-40MnMo43	SCCrM1	25ChGM	14 2225	25CD45	—
C 0.28/0.33 Mn 0.4/0.6 Cr 0.8/1.0 Mo 0.15/0.25	4130 (G41300)	708A30	—	SCM2 SCM430	30ChMA	14 2233	30CD4	30CrMo
C 0.33/0.35 Mn 0.7/0.9 Cr 0.6/1.10 Mo 0.15/0.25	4135 (G41350)	708A37 (19B)	CS-34CrMo4 (1.7220)	SCM3H	35ChM	14 2234	35CD4	35CrMo
C 0.38/0.43 Mn 0.75/1.00 Cr 0.8/1.10 Mo 0.15/0.25	4140 (G41400)	708A40 (19A)	42CrMo4 (1.7225)	SCM4	38ChM	14 2244	42CD4	42CrMo
C 0.38/0.43 Mn 0.6/0.8 Cr 0.7/0.9 Ni 1.65/2.00 Mo 0.2/0.3	4340 (G43400)	816M40 (110)	40NiCrMo6 (1.6565)	SNCM8	40ChMA	14 2541	35NCD6	—
C 0.18/0.23 Mn 0.45/0.65 Mo 0.45/0.60	4419 (G44190)	—	GS-22Mo4 (1.5419)	SCPH11	—	—	18MD4.05	—
C 0.13/0.18 Mn 0.7/0.9 Cr 0.7/0.9	5115 (G51150)	527A17	16MnCr5 (1.7131)	SCr21	15Ch	14 2127	16MC5	15Cr
C 0.17/0.22 Mn 0.7/0.9 Cr 0.7/0.9	5120 (G51200)	—	20CrMnS3 (1.7121)	SCr22	20Ch	—	20MC5	20Cr
C 0.28/0.33 Mn 0.7/0.9 Cr 0.8/1.1	5130 (G51300)	530A30 (18A)	30MnCrTi4 (1.8401)	SCr2H	27ChGR	—	28C4	30Cr

C 0.33/0.38 Mn 0.6/0.8 Cr 0.8/1.05	5135 (G51350)	530A36 (18C)	38Cr4 (1.7043)	SCr3H	35Ch	—	38C4	35Cr
C 0.38/0.43 Mn 0.7/0.9 Cr 0.7/0.9	5140 (G51400)	530A40 (18D)	42Cr4 (1.7045)	SCr4H	40Ch	14 2245	42C4	40Cr
C 0.98/1.10 Mn 0.25/0.45 Cr 1.3/1.6	E52100 (G52986)	535A99 (31)	105Cr5 (1.2060)	SCr5	SchCh15	14 2258	100C6	—
C 0.48/0.53 Mn 0.7/0.9 Cr 0.8/1.1 V 0.15	6150 (G61500)	735A50 (47)	GS-50CrV4 (1.8159)	SUP10	50ChF	14 2230	50CV4	50CrVA
C 0.18/0.23 Mn 0.7/0.9 Cr 0.4/0.6 Ni 0.4/0.7 Mo 0.15/0.25	8620 (G86200)	805A20	21NiCrMo2 (1.6523)	SNCM21	AS20ChGNM	14 2506-03	20NCD2	20CrNiMo
C 0.38/0.43 Mn 0.75/1.0 Cr 0.4/0.6 Ni 0.4/0.7 Mo 0.15/0.25	8640 (G86400)	945A40 (10C)	40NiCrMo2 (1.6546)	SNCM6	38ChGNM	—	40NCD2	—
1.1.4 Stainless steels								
C 0.9 Cr 16/18 Ni 6.5/7.75 Al 0.75/1.5	17-7 PH (S17700)	301S81	X7CrNiAl17 7 (1.4564)	SUS631	—	—	177F00	1Cr17Ni7Al

(*continued*)

Table 1.1 RELATED DESIGNATIONS FOR STEELS—*continued*

Nominal composition	*USA AISI/SAE (UNS)*	*UK BS 970 (En)*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Sweden SIS*	*France AFNOR*	*China GB*
C 0.15 Mn 7.5/10.0 Cr 17/19 Ni 4/6 N 0.25	202 (S20200)	284S16	X8CrMnNi18 8 (1.3965)	SUS202	1Ch17N3G8AE	14 2357	—	1Cr18Mn8Ni5N
C 0.15 Mn 2.0 Cr 17/18 Ni 8/10	302 (S30200)	302S25 (58A)	X12CrNi18 8 (1.4300)	SUS302	Ch18N9	14 2331	302F00 Z10CN18.9	1Cr18Ni9
C 0.8 Mn 2.0 Cr 18/20 Ni 8/10.5	304 (S30400)	304S15 (58E)	X5CrNi18 10 (1.4301)	SUS304	0Ch18N10	14 2332	304F01 Z5CN18.09	0Cr18Ni9
C 0.03 max Mn 2.0 Cr 18/20 Ni 8/12	304L (S30403)	304S11 (58E)	X2CrNi19 11 (1.4306)	SUS304L	03Ch18N11	14 2352	304F11 Z2CN18.10	00Cr18Ni10
C 0.25 Mn 2.0 Cr 24/26 Ni 19/22	310 (S31000)	310S24	X12CrNi25 21 (1.4845)	SUSY310	Ch25N20	—	310F00 Z12CN25.20	2Cr25Ni20
C 0.8 Mn 2.0 Cr 16/18 Ni 10/14 Mo 2/3	316 (S31600)	316S16 (58J)	X15CrNiMo17 13 3 (1.4436)	SUS316	SW-04Ch19Ni1M3	14 2343	316F00	0Cr17Ni20Mo2
C 0.03 max Mn 2.0 Cr 16/18 Ni 10/14 Mo 2/3	316L (S31603)	316	X2CrNiMo18 14 3 (1.4435)	SUSY316L	03Ch16N15M3	14 2348	22CND17.12	00Cr17Ni14Mo2

C 0.08 Mn 2.0 Cr 16/18 Ni 10/14 Mo 2/3 Ti 5x(C+N)/0.7	316Ti (S31635)	320S17 (58J)	X6CrNiMoTi17 12 2 (1.4571)	—	08Ch17N13M2T	14 2350	Z6CNDT17.12	0Cr18Ni12Mo2Ti
C 0.08 Mn 2.0 Cr 17/19 Ni 10/14 Mo 2/3 Nb 10xC/1.10	316Cb (S31640)	318C17	X6CrNiMoNb17 12 2 (1.4580)	SCS22	0Ch18N9MB	—	Z4CNDNb18.12M	—
C 0.08 Mn 2.0 Cr 17/19 Ni 9/12 Ti 5xC min	321 (S32100)	321S20 (58B/C)	X6CrNiTi18 10 (1.4541)	SUS321	08Ch18N10T	14 2337	Z6CNT18.10	0Cr18Ni11Ti
C 0.08 Mn 2.0 Cr 17/19 Ni 9/13 Nb+Ta 10xC min	347 (S34700)	347S31	X5CrNiNb18 10 (1.4546)	SUS347	Ch18N11B	14 2338	Z4CNNb19.10M	0Cr18Ni11Nb
C 0.15 Mn 1.0 Cr 11.5/13.0	403 (S40300)	410S21 (56A)	X10Cr12 (1.4006)	SUS403	15Ch13L	14 2301	Z15C13	1Cr12
C 0.15 Mn 1.0 Cr 11.5/13.5 Si 1.0	410 (S41000)	—	X15Cr13 (1.4024)	SUS416	12Ch13	14 2302	Z12C13M	1Cr13
C 0.15 min Mn 1.0 Cr 12/14 Si 1.0	420 (S42000)	420S37 (56C)	X20Cr13 (1.4021)	SUS420J2	20Ch13	14 2304	Z20C13N	2Cr13

(*continued*)

Table 1.1 RELATED DESIGNATIONS FOR STEELS—*continued*

Nominal composition	*USA AISI/SAE (UNS)*	*UK BS 970 (En)*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Sweden SIS*	*France AFNOR*	*China GB*
C 0.12 Mn 1.0 Cr 16/18	430 (S43000)	430S15 (60)	X8Cr17 (1.4016)	SUS430	12Ch17	14 2320	Z8C17	1Cr17
C 0.48/0.58 Mn 8/10 Cr 20/22 Ni 3.25/4.5 N 0.28/0.5	21-4N (S63008)	349S52	X5CrMnNiN21 9 (1.4871)	SUH36	5Ch20N4AGN	—	Z52CMN21.09	Y5Cr21Mn9Ni4N
1.1.5 Tool steels								
C 0.8/1.0 Cr 3.75/4.5 Mo 4.5/5.5 W 5.5/6.75 V 1.75/2.2	M2 (T11302rc)	BM2	S6-5-2 (1.3343)	SKH51	R6AM5	14 2722	6-5-2	W6Mo5Cr4V2
C 0.65/0.8 Cr 3.75/4.5 W 17.25/18.75 V 0.9/1.3	T1 (T12001)	BT1	S18-0-1 (1.3355)	SKH2	R18	—	18-0-1	W18Cr4V
C 1.5/1.6 Cr 3.75/5.0 Mo 1.0 max W 11.75/13.0 V 4.5/5.25 Co 4.75/5.25	T15 (T12015)	BT15	S12-1-4-5 (1.3202)	SKH10	R10K5F5	—	12-1-5-5	—
C 0.35/0.45 Si 0.8/1.2 Cr 3.0/3.75 Mo 2.0/3.0 V 0.25/0.75	H10 (T20810)	VH10	X32CrMoV3 3 (1.2365)	SKD7	3Ch3M3F	—	32DCV28	—

C 0.33/0.43 Si 0.8/1.2 Cr 4.75/5.0 Mo 1.1/1.6 V 0.3/0.6	H11 (T20811)	BH11	X41CrMoV5 1 (1.7783)	SKD6	4Ch5MF5	—	FZ38CDV5	—
C 0.32/0.45 Si 0.8/1.2 Cr 4.75/5.5 Mo 1.1/1.75 V 0.8/1.2	H13 (T20813)	BH13	X40CrMoV5 1 (1.2344)	SKD61	4Ch5MF15	14 2242	Z40CDV5	—
C 0.26/0.36 Cr 3.0/3.75 W 8.5/10.0 V 0.3/0.6	H21 (T20821)	BH21	X30WCrV9 3 (1.2581)	SKD5	3Ch2W8F	14 2730	Z30WCV9	—
C 1.4/1.6 Cr 11.0/13.0 Mo 0.7/1.2 V 1.1 max	D2 (T30402)	BD2	X155CrMo12 1 (1.2379)	SKD11	—	14 2310	Z200C12	3-2 Cr12MoV
C 2.0/2.25 Cr 11.0/13.5 W 1.0 max V 1.0 max	D3 (T30403)	BD3	X210CrW12 (1.2436)	SKD1	Ch12	—	Z200C12	3-1 Cr12
C 0.4/0.55 Si 0.15/1.2 Cr 1.0/1.8 W 1.5/3.0 V 0.15/0.30	S1 (T41901)	BS1	45WCrV7 (1.2436)	SKS41	4ChW2S	14 2710	55WC20	2-2 5CrW2Si
C 0.7/1.5	W1 (T72301)	BW1B	C125W1 (1.1663)	SK2	U12	14 1880	Y(2) 120	3 85
C 0.8/1.5 V 0.15/0.35	W2 (T72302)	BW2	100V1 (1.2833)	SKS43	—	—	Y105V	1-8V

Table 1.2 RELATED DESIGNATIONS FOR WROUGHT ALUMINIUM ALLOYS

UNS	*ISO—Nominal composition*	*USA/Japan*	*UK Former BS*	*Germany DIN (Wk. No.)*	*Russia GOST*	*Sweden SIS*	*France Former NF*	*China GB*
A91050	Al99.5	1050	1B	Al99.5 (3.0255)	Al	4007	A5	L4
A91070	Al99.7	1070		Al99.7 (3.0275)	A7	4005	A7	L2
A91080	Al99.8	1080	1A	Al99.8 (3.0285)	—	4004	A8	L1
A91100	Al99.0Cu	1100	—	—	A2		A45	—
A91145	—	1145	—	—	AE, AT	—	—	—
A91200	Al99.0	1200	1C	Al99.0 (3.0205)	A2	4010	A4	—
A91350	—	1350	1E, E57S	EAl99.5 (3.0257)	A00	4008	—	—
A92011	Al-Cu6BiPb	2011	FC1	AlCuBiPb (3.1655)	—	4355	A-U5PbBi	—
A92014	Al-Cu4SiMg	2014	H15	AlCuSiMn (3.1255)	1380, AK8	4338	A-U4SG	—
A92017	Al-Cu4MgSi	2017	—	AlCuMg1 (3.1325)	AK10, D17	—	A-U4G	LD9, LD10
A92024	Al-Cu4Mg1	2024	2L97, 2L98, L109, L110	AlCuMg2 (3.1355)	1160, D16	—	A-U4G1	LY6, LY9, LY12
A92031	Al-Cu2NiMgFeSi	2031	H12	—	—	—	A-U2N	—
A92117	Al-Cu2Mg	2117	3L86	AlCu2.5Mg0.5 (3.1305)	1180, D18	—	A-U2G	—
A92219		2219		—	1201			
A92618	Al-Cu2Mg1.5Fe1Ni1	2618	H16	—	1141, AK4-1	—	—	—
A93003	Al-Mn1Cu	3003	N3	AlMnCu (3.0517)	1400, AMts	4054	A-M1	—
A93004	—	3004	—	AlMn1Mg1 (3.0528)	AMts2	—	A-M1G	—
A93103	Al-Mn1	3103	N3	AlCu2.5Mg0.5 (3.1305)	D18	—	—	—
A93105	Al-MnMg	3105	N31, E4S	AlMn0.5Mg0.5 (3.0505)	—	—	—	—
A94032	—	4032	38S	—	—	—	—	—
A94043		4043	N21	S-AlSi5 (3.2245)	—	4225	—	LT1
A94047	Al-Si12	4047	N21	—	—	—	A-S12	—
A95005	Al-Mg1	5005	N41	AlMg1 (3.3315)	1510, AMg1	4106	A-G0.6	—

A95050	Al-Mg1.5	5050	3L44	AlMg1.5 (3.3318)	—	—	A-G1.5	—
A95052	Al-Mg2.5	5052	2L55	AlMg2.5 (3.3523)	1520, AMg2	4120	A-G2.5C	LF2
A95056	Al-Mg5	5056	N6, A56S	AlMg5 (3.3555)	AMg5	4146	—	LF10, LF5-1, LF51
A95083	Al-Mg4.5Mn	5083	N8	AlMg4.5Mn (3.3547)	—	4140	A-G.5MC	LF4
A95086	Al-Mg4	5086		AlMg4.5 (3.3345)	1540, AMg4	—	A-G4MC	—
A95154	Al-Mg3.5	5154	N5	—	1530, AMg3	—	AG3	—
A95251	Al-Mg2	5251	N4	AlMg2Mn0.3 (3.3525)	—	—	A-G2M	—
A95454	Al-Mg3Mn	5454	N51	AlMg2.7Mn (3.3537)	—	—	A-G2.5MC	—
A95456	—	5456	—	—	45Mg2, AMg6	—	A-G5MC	—
A95754	Al-Mg3	5754	—	AlMg3 (3.3525)	1530, AMg3	4133	A-G3M	—
A96060	—	6060	—	AlMgSi0.5 (3.3206)	—	4103	A-GS	—
A96061	Al-Mg1SiCu	6061	H20, E91E	AlMgSiCu (3.3211)	1330, AD33		A-GSUC	
A96063	Al-MgSi	6063	H9	—	1310, AD31	4104	—	—
A96082	Al-Si1MgMn	6082	H30	AlMgSi1 (3.2315)	—	4212	A-SGM0.7	—
A96101	—	6101	91E	E-AlMgSi0.5 (3.3207)	1310, AD31	4102	A-GS/L	—
A96463	—	6463	BTR6	Al99.85MgSi (3.2307)			A85-GS	—
A97005	—	7005		AlZnMgCu0.5 (3.4345)	1910,m ATsM	4212	—	—
A97020	Al-Zn4.5Mg	7020	H17	(3.4335)		4425	A-Z5G	—
A97072	Al-Zn1	7072	—	AlZn1 (3.4415)	AtsP1	—	—	LB1
A97075	Al-Zn6MgCu	7075	2L95, L160, L161, L162, C77S	AlZnMgCu1.5 (3.4365)	1950, V95	—	A-Z5GU	LC4, LC9

Table 1.3 RELATED DESIGNATIONS FOR COPPER ALLOYS

UNS	*ISO—Nominal composition*	*USA*	*UK Former BS*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Sweden SIS*	*France Former NF*	*China GB*
1.3.1 Coppers									
C10200	Cu-OF	OF	C103	SE-Cu (2.0070)	C1020	V3	5011	Cu/c1, Cu/c2	Cu-1, Cu-2
C11000	Cu-ETP, Cu-FRHC	ETP	C101	E-Cu57 (2.0060)	C1100	—	5010	Cu/a1, Cu/a2	T2
C11600	CuAg0.1	STP		CuAg0.1 (2.1203)		—	5030	—	TAg0.1
C12200	Cu-DHP	DHP	C106	SF-Cu (2.0090)	C1220	—	5105		—
C12500	Cu-FRTP	FRTP	C104	(2.008)	—	—	5013	Cu/a3	—
C14200	CuAs(P)	DPA	C107	CuAs(P) (2.0150)	—	—	—	—	—
C14500	Cu-Te, Cu-Te(P)	DPTE	C109	CuTeP (2.1646)	—	—	—	—	—
C16200	CuCd1	CDA 162	C108	CuCd1 (2.1266)	—	MTsBB	5055	—	QCd1
C17000	CuBe1.7	CDA 170	CB101	CuBe1.7 (2.1245)	C1700	BHT 1.7	—	CuBe1.7	QBe1.7
C17200	CuBe2	CDA 172		CuBe2 (2.1247)	C1720	BHT 1.9, M2, M3	—	—	QBe2
C17500	CuCo2Be	CDA 175	A 3/1	CuCo2Be (2.1285)	—	—	—	—	—
C18200	CuCr1	CDA 182	A 2/1	CuCr (2.1291)	—	Br.Kh, Br.Kh.5, Br.Kh.7, Br.Kh.8	—	—	—
1.3.2 Brasses									
C21000	CuZn5	CDA 210	CZ125	CuZn5 (2.0220)	C2100	L96, LT96	—	U-Z5	H96
C22000	CuZn10	CDA 220	CZ101	CuZn10 (2.0230)	C2200	—	—	U-Z10	H90
C23000	CuZn15	CDA 230	CZ102	CuZn15 (2.0240)	C2300	L85, L87	5112	U-Z15	H85
C24000	CuZn20	CDA 240	CZ103	CuZn20 (2.0250)	C2400	L80	5114	U-Z20	H80
C26000	CuZn30	CDA 260	CZ106	CuZn30 (2.0285)	C2600	L69, L70, LMSH68-.06	5122	U-Z30	H70
C26800	CuZn33	CDA 268	CZ107	CuZn33 (2.0280)	C2680	L66, L68	—	U-Z33	—
C27400	CuZn37	CDA 274	CZ108	CuZn37 (2.0320)	C2740	L63	5150	U-Z37	H62, H63
C28000	CuZn40	CDA 280	CZ109	CuZn40 (2.0360)	C2800	L58, L60	5163	U-Z40	H59
C35300	CuZn35Pb2	CDA 353	CZ119	CuZn36Pb1.5 (2.0331)	—	LS64-2	5165	—	—
C36000	CuZn36Pb3	CDA 360	CZ124	CuZn36Pb3 (2.0375)	C3603	LS63-3	—	U-Z36Pb3	HPb63-3
C37700	CuZn38Pb2	CDA 377	CZ120	CuZn38Pb2 (2.0380)	C3561	LS59-1, LS59-1L, LS59-1V	5168	—	—
C38000	CuZn43Pb2	CDA 380	CZ122	CuZn43Pb2 (2.0402)	—	—	5168	U-Z39Pb2	—
C38500	CuZn39Pb3	CDA 385	CZ121	CuZn39Pb3 (2.0401)	—	—	5170	—	—
C44300	CuZn28Sn	CDA 443	CZ111	CuZn28Sn (2.0470)	—	LO70-1, LOAs70-1-005	5220	U-Z29E1	HSn70-1
C46400	CuZn38Sn1	CDA 464	CZ112	CuZn38Sn (2.0530)	C4641	LO60-1, LO62-1	—	—	HSn62-1

1.3.3 Phosphor bronzes

C51000	CuSn4	CDA 510	PB102	—	C5101	Br.OF5.5-.3	—	—	—
C51100	CuSn4	CDA 511	PB101	—	C5101	Br.OF4-.25	—	—	QSn4-0.3
C51900	CuSn6	CDA 519	PB103	CuSn6 (2.1020)	C5191	Br.OF6.5-.15	5428	U-E5P	QSn6.5-0.1 QSn6.5-0.4
C52100	CuSn8	CDA 521	PB104	CuSn8 (2.1030)	—	Br.OF7-.2 Br.OF7-.4	—	U-E9P	QSn7-0.2
C52400	CuSn10	CDA 524	—	—	—	Br.O10-1 Br.OF10-1	—	U-E9P	—
C54400	CuSn4Pb4Zn3	CDA 544	—	—	C5441	Br.OSTs4-4-.25 Br.OSTs4-4-2.5	—	—	QSn4-3 QSn4-4-2.5 QSn4-4-4

1.3.4 Aluminium-silicon bronzes

C60800	CuAl5	CDA 608	CA101	CuAl5As (2.0918)	—	BRA5	—	U-A6	—
C62300	CuAl10Fe3	CDA 623	CA103	CuAl10Fe (2.0938)	C6161	Br.VAZh	—	—	QAl9-4
C63000	CuAl10Fe5Ni5	CDA 630	CA105	CuAl11Fe6Ni5 (2.0978)	C6031	Br.AZhN10-4-4L Br.AZhN9-4-4	—	—	QAl/10-4-4 QA/11-6-6
C65500	CuSi3Mn1	CDA 655	CS101	CuSi3Mn (2.1525)	—	Br.KMTs3-1	—	—	QSi3-1
C68700	CuZn20Al2	CDA 687	CZ110	CuZn20Al (2.0460)	—	LA77-2	—	U-Z22A2	—

1.3.5 Copper-nickel and nickel silvers

C70600	CuNi10Fe1Mn	CDA 706	CN102	CuNi10Fe1 (2.0872)	C7060	MNZhMTs10-1.5-1	5667	U-N10	BFe10-1-1
C71000	CuNi20Mn1Fe	CDA 710	CN104, NS108, NS109	CuNi20Fe (2.0878)	C7100	—	—	—	—
C71500	CuNi30Mn1Fe	CDA 715	CN106, CN107	CuNi30Fe (2.0882)	C7150	HM70, MH33	5682	U-N30	BFe30-1-1
C75200	CuNi18Zn20	CDA 752	NS106	CuNi18Zn (2.0740)	C7521	MNTs	5246	U-Z22N18	—
C75700	CuNi12Zn24	CDA 757	NS104	CuNi12Zn24 (2.0730)	—	—	5243	—	—
C77000	CuNi18Zn27	CDA 770	NS107	—	C7701	—	—	U-Z27N18	—

Table 1.4 RELATED DESIGNATIONS FOR MAGNESIUM ALLOYS

UNS	*ISO—Nominal composition*	*USA ASTM*	*UK BS 2970*	*Germany DIN (Wk. No.)*	*Japan JIS*	*Russia GOST*	*Europe AECMA*	*France AFNOR*
1.4.1 Cast alloys								
M11101	Mg-Al9Zn	AZ101A	MAG 3	—	—	MGS6, ML6	—	—
M11810	Mg-Al8Zn	AZ81A	MAG 1	G-MgAl8Zn1 (3.5812)	MB2, MS2	ML5	MG-C-61	G-A9
M11910	Mg-Al9Zn	AZ91A	—	GD-MgAl9Zn (3.5912)	MDC1A	—	—	—
M11914	—	AZ91C	MAG 3	G-MgAl9Zn (3.5912)	MC2	—	—	G-A9Z1
M12330	Mg-RE3Zn2Zr	EZ33A	MAG 6	G-MgSE3Zn2Zr1 (3.5103)	MC8	—	MG-C-91	G-TR3Z2
M13320	Mg-Th3Zn2Zr	HZ32A	MAG 8	MgTh3Zn2Zr1 (3.5105)	—	ML14	MG-C-81	G-Th3Z2
M16410	Mg-Zn4REZr	ZE41A	MAG 5	MgZn4SE1Zr1 (3.5101)	—	—	MG-C-43	G-Z4TR
M16510	Mg-Zn5Zr	ZK51A	—	—	MC6	ML12, VM65	—	—
M16610	Mg-Zn6Zr	ZK61A	—	—	MC7	VM65-3	—	—
M19995	Mg99.95	9995A	—	H-Mg99.95 (3.5002)	—	—	—	—
1.4.2 Wrought alloys								
M11311	Mg-Al3Zn1	AZ31B	MAG S-1110	MgAl3Zn (3.5312)	MT1, MB1, MS1	—	Mg-P-62	G-A3Z1
M11600	Mg-Al6Zn1	AZ61A	—	MgAl6Zn (3.5612)	MT2, MB2, MS2	MA3	Mg-P-63	G-A6Z1
M11800	Mg-Al8Zn	AZ80A	MAG 7	MgAl8Zn (3.5812)	MB3, MS3	MA5	Mg-P-61	G-A8Z1

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